

Nga Hoang

AI ENGINEER · SOFTWARE ENGINEER · PYTHON DEVELOPER · SOLUTION ARCHITECT

Tang Bat Ho Street, Hai Chau District, Da Nang City

☎ (+84) 794-508-421 | ✉ ngahv2222@gmail.com | 🏠 ngahoangdev.com | 📱 [ngahoangvan](https://www.instagram.com/ngahoangvan) | 🌐 [ngahoangvan](https://www.linkedin.com/company/ngahoangvan)



Summary

Software engineer with 7+ years of experience building production systems across AI integration, backend development, and cloud architecture. Core expertise in Python (Django, FastAPI, Flask) with hands-on experience shipping LangChain/OpenAI-powered features, real-time data pipelines, and microservices at scale. Proven ability to lead small teams, architect end-to-end solutions, and deliver measurable business impact — from 30% faster processing pipelines to trading platforms saving users 40% of their time. Experienced across AWS, GCP, and Azure with a strong track record in technical solution design and cross-functional collaboration with international clients.

Skills

- Python | JavaScript | TypeScript | SQL | Django | FastAPI | Flask | Dash/Plotly | NodeJS | Express | React | AngularJS | NestJS
- AI Integration: LangChain | Tensorflow | Pytorch | Keras | Sklearn | OpenAI | RAG | Firecrawl
- Database: PostgreSQL | MySQL | MongoDB | Redis | TimescaleDB | Snowflake | Vector Database | Elasticsearch
- AWS | GCP | Azure | Docker | Kubernetes | Terraform | Ansible | CI/CD | Airflow | ELK | Prometheus | Grafana | Kafka | Celery
- Microservices | Distributed Systems | TDD | Scrum | Full-stack | Backend | Frontend

Experience

Saigon Technology

Da Nang, Vietnam

SOFTWARE ENGINEER / AI ENGINEER

Aug 2022 – Present

- Built an AI-powered web scraping service using Firecrawl that extracts structured data from arbitrary URLs, backed by background jobs and batch processing on AWS Lambda, S3, and SES.
- Designed and deployed ETL pipelines that feed into a **LangChain/RAG/OpenAI** engine to predict career pathways and recommend courses — orchestrated end-to-end via AWS Step Functions and Batch.
- Architected a **real-time AI chatbot** widget using FastAPI + Websocket, integrating GPT-3.5/GPT-4 via LangChain to handle live customer conversations.
- Automated data syncing with Airtable using Celery workers and Redis caching, reducing manual intervention and improving data freshness across the platform.
- Owned the Airflow-based data pipeline for a US banking client — built DAGs for multi-source ingestion and report generation, migrated to the latest Airflow release, and integrated LangChain-powered microservices for AI-enhanced data flow.
- Developed a face recognition and person tracking system with Django + LangChain; built an OpenAI-powered chatbot for automated report summarization.
- Achieved **30% reduction in task processing time** through parallel processing optimizations. Stood up ELK, Kafka, Apache Superset, and Bitbucket CI/CD pipelines from scratch.
- Technologies:
 - **Language:** Python, TypeScript, JavaScript, Bash
 - **Framework:** FastAPI, Django, Flask, NestJS, ReactJS
 - **AI:** LangChain, OpenAI, RAG, Firecrawl
 - **Tools:** Airflow, Celery, Kafka, ELK, Apache Superset, Docker
 - **Database:** PostgreSQL, Redis, Snowflake, Vector DB
 - **Cloud:** AWS (S3, Lambda, SES, ECS, Step Function, Batch, ECR)
 - **CI/CD:** Bitbucket Pipeline, AWS CodeBuild, AWS CodePipeline

Datatize

Singapore

SOFTWARE ENGINEER & QUANTITATIVE DEVELOPER

May 2021 – Mar 2024

- **Led a team of 3 developers** to build and ship a live crypto trading platform serving multiple exchanges (Binance, FTX, Deribit), **saving users up to 40% of trading time** through automated strategy execution.
- Built an AI-assisted developer toolchain using LangChain + OpenAI for SQL generation, code generation, and automated code vulnerability scanning — adopted across the engineering team.
- Engineered a real-time Data Visualization System with Dash/Plotly and ReactJS, integrating LangChain for AI-driven analytics and Sklearn/LinearRegression for price prediction models.
- Developed Airflow DAGs to crawl and process data from multiple crypto exchanges. Built a cross-platform mobile app with React Native alongside the web product.
- Stood up the full DevOps stack: CI/CD via GitHub Actions, centralized monitoring with ELK + Prometheus + Grafana, and Tableau for business intelligence.
- Technologies:
 - **Language:** Python, JavaScript, Bash
 - **Framework:** Dash/Plotly, ReactJS, React Native, Jinja2
 - **AI:** LangChain, OpenAI, Sklearn
 - **Tools:** Airflow, ELK, Prometheus, Grafana, Tableau, Docker
 - **Database:** PostgreSQL - TimescaleDB, MongoDB, Redis
 - **Cloud:** Firebase, Google Cloud Platform
 - **CI/CD:** GitHub Action

STI

Da Nang, Vietnam

DEVOPS

Dec 2021 – Mar 2022

- Provisioned the entire AWS infrastructure for an NFT Marketplace using Terraform (S3, SES, Route53, VPC), enabling reproducible and version-controlled deployments.
- Built a fully automated CI/CD pipeline with GitLab CI + Terraform, eliminating manual deployment steps and reducing release cycles.
- Technologies:
 - **Language:** Bash
 - **Infra as Code:** Terraform
 - **CI/CD:** GitLab CI
 - **Cloud:** AWS

Crypto Subscriptions

United States

FREELANCER FULL-STACK DEVELOPER

Dec 2021 – Aug 2022

- **Sole developer** who built a full-stack crypto subscription platform end-to-end — Django REST API backend, ReactJS frontend, KYC/MFA security layer, and payment integration.
- Integrated Persona and Sumsb for KYC, Twilio Authy for MFA, and Wyre + Plaid for payment processing, meeting US financial compliance requirements.
- Designed a cron-based subscription lifecycle engine handling recurring billing, renewals, and expiration automatically.
- Technologies:
 - **Backend:** Python, Django, Django Rest Framework
 - **Frontend:** JavaScript, ReactJS, HTML, CSS
 - **Database:** PostgreSQL
 - **Integrations:** Twilio Authy, Persona, Sumsb, Wyre, Plaid

D-soft, JSC.

Da Nang, Vietnam

SOFTWARE ENGINEER

Mar 2019 – Apr 2021

- Built the Flask backend for a Face Recognition System and a Django-based Debugging Tool, both deployed on Azure Kubernetes Service with fully containerized workflows.
- Established GitHub Actions CI/CD pipelines for automated testing and deployment to AKS, standardizing the team's release process.
- Technologies:
 - **Language:** Python, JavaScript, Bash
 - **Framework:** Flask, Django
 - **Cloud:** Azure Kubernetes Service, Azure Container Registry
 - **CI/CD:** GitHub Action, Docker

GMO-Z VietNamLab Center

Da Nang, Vietnam

INTERNSHIP

Jan 2019 – Mar 2019

- Developed a web-based Japanese Examination System and automated infrastructure provisioning with Ansible + Docker.
- **Technologies:** PHP, Yii2, Ansible, Docker.

Certificates

- 2023 **AWS Certified Solutions Architect – Associate**, Amazon Web Services (AWS)
- 2023 **Data Visualization with Python**, Coursera & IBM
- 2023 **Data Analysis with Python**, Coursera & IBM

Education

